

Welcome to RASCAL!

WebGL User Manual

Version 1.0



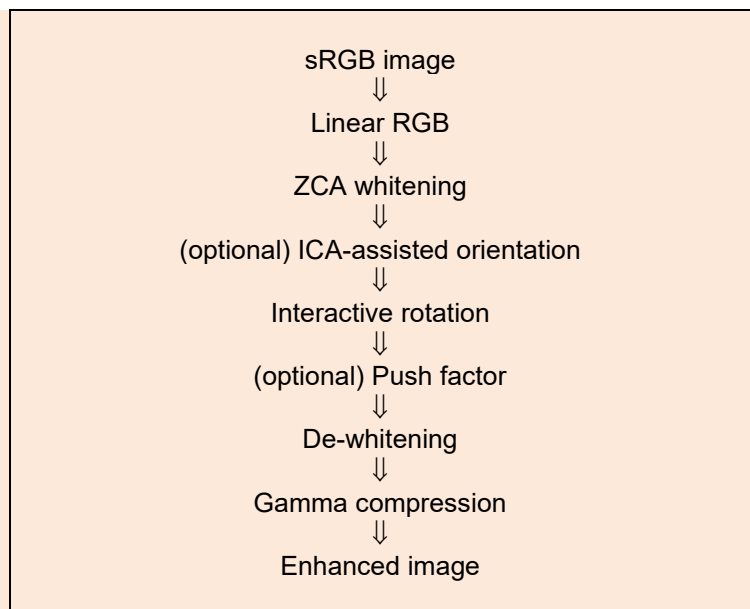
Edition	WebGL
Platform	Windows, macOS , Linux , Android , iOS, and iPadOS, provided that the browser supports WebGL2

1. Presentation

RASCAL (**R**adial **A**ngular **S**tatistical **C**olor **A**daptation and **L**ifting) is an interactive software designed to reveal subtle chromatic variations in standard RGB digital images. The program was developed primarily for applications in cultural heritage and archaeological imaging, where faint color differences may correspond to pigments, surface alterations, or archaeological structures that are difficult to detect visually. The application performs a decorrelation of RGB color channels followed by an interactive exploration of color transformations. This approach allows the user to visually enhance color contrasts and to concentrate relevant chromatic information on one or several of the image channels. Unlike fully automated enhancement techniques, RASCAL relies on seamless interactive exploration. The operator visually evaluates the transformations and identifies the most informative configuration.

2. Conceptual overview

Digital images are typically encoded in the sRGB color space, where the three-color channels may be strongly correlated. Such correlations often confine color information to a restricted region of the RGB cube, which reduces the visibility of some chromatic variations. RASCAL addresses this limitation using the following workflow:



After whitening in the linear RGB color space, the covariance matrix of the color distribution becomes spherical, meaning that all chromatic directions have unit variance. Because such a distribution is invariant under orthogonal rotations, the user can rotate the color space to align chromatic variations of interest with the RGB axes. Note that the whitening transformation is computed once from the original image (or from a subset of it) and remains fixed during the entire exploration. ICA-assisted orientation can also be chosen as starting point, as it may help separate statistically independent chromatic components. In any case, subsequent rotations do not modify the statistical model. As a result, the rotation performed by RASCAL does not correspond to a simple rotation of the RGB axes. Instead, the rotation operates in the whitened color space, whose geometry depends on the covariance structure of the image. Consequently, the transformation is

image-dependent and reflects the internal geometry of the color distribution. This process often reveals structures that are difficult or impossible to perceive in the original image. After rotation, the transformed data are mapped back to the original RGB space through the inverse whitening (de-whitening) transform, restoring the original variance structure. The resulting linear RGB values are then converted back to sRGB through gamma compression for display.

The detailed mathematical description of the method is provided in the “Materials and Methods” section of the associated article.

3. Technical overview

RASCAL WebGL is implemented as a self-contained single-page web application written in standard HTML, CSS, and JavaScript. It runs entirely in the user’s browser and does not require installation. The application requires a recent browser with WebGL2 support.

4. Loading an image

Images are opened through the browser file interface and are processed locally; no image data are uploaded. RASCAL WebGL accepts common image formats that can be decoded directly by the web browser, such as JPEG, PNG, and WebP. Since image decoding is handled by the browser, support for some additional formats may vary between systems and browsers. TIFF and RAW files are not supported in the WebGL version and should be converted beforehand, preferably to PNG or to a high-quality JPEG.

To keep the application responsive and to avoid excessive memory use in the browser, very large images are automatically resized at loading time. If the longest side of the input image exceeds 4096 pixels, the image is downscaled proportionally so that its largest dimension becomes 4096 pixels, while preserving the original aspect ratio. The displayed filename indicates both the original and working dimensions. All subsequent operations, including ZCA computation, ROI processing, ICA-assisted orientation, interactive rendering, presets, and export, are then performed on this working image. Consequently, exported images are saved at the working resolution, not necessarily at the original full resolution.

TIP: RASCAL WebGL generally gives better results on sufficiently bright images. Dark images may benefit from brightness, exposure, or contrast adjustment before processing.

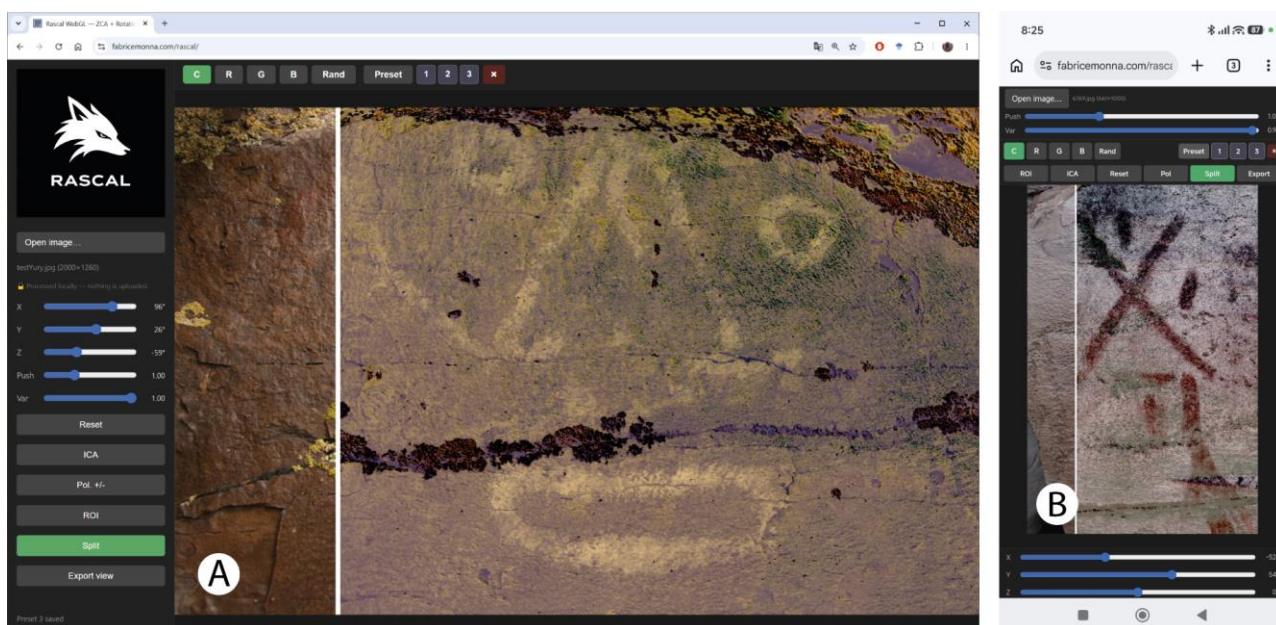


Figure 1. Responsive interface layouts of RASCAL WebGL. (A) Desktop layout. (B) Mobile layout.

5. Interface description

RASCAL WebGL uses a responsive interface that automatically adapts to the size and orientation of the screen (Figure 1). On desktop and large tablet displays, the controls are distributed between a lateral toolbar, containing image loading and the main adjustment sliders, and a top toolbar, containing display modes, presets, and action buttons. On smaller screens, especially smartphones, the layout is reorganized to preserve the largest possible image display area: the most frequently used tools are moved to compact top bars, while the rotation sliders are placed at the bottom of the screen for easier touch interaction. The available functions remain essentially the same in both layouts, but their position and grouping may differ slightly depending on the device.

6. Rotation and navigation

The user can interactively explore the transformed color space either through the dedicated sliders or through the X, Y, and Z sliders with the preview canvas. The available gestures depend on the device. On desktop computers, the image can be controlled with the mouse:

Left mouse drag → rotation around the Z axis

Right mouse drag → rotation around the Y axis

Mouse wheel → rotation around the X axis

Ctrl + mouse wheel / ⌘ + mouse wheel on macOS → zoom in/out, centered on the pointer position

On touch-screen devices, the image can be controlled directly with the fingers:

One-finger drag → horizontal movement rotates around the Y axis, vertical movement rotates around the X axis

Two-finger twist → rotation around the Z axis

Two-finger pinch → zoom in/out, centered on the midpoint between the fingers

Double tap → reset the current transformation, except in ROI mode where it validates the painted ROI

Clicking or tapping the **Random** button (or pressing the **Ctrl + Shift + R** keys on desktop) launches an automatic sequence of random color space orientations. A new random orientation is generated every 3 seconds, and the X, Y and Z sliders are updated accordingly. The sequence continues until the user interacts with any control, including a slider, mouse drag, mouse wheel movement, or finger drag, at which point it stops immediately. Pressing Random again starts a new sequence.

Clicking or tapping the **Reset** button (or pressing the **R** key on desktop) restores all rotation angles to zero, returns the Push factor and Variance restoration controls to their default values, clears the zoom, and restores the current ZCA whitening state.

7. Polarity inversion

Clicking or tapping the **Pol. +/-** button reverses the signs of the three Euler rotation angles used in the ZCA color space. In practice, the current rotation angles X, Y, and Z are replaced by $-X$, $-Y$, and $-Z$, and the rotation matrix is immediately recomputed from these new values. This provides a quick way to explore an alternative polarity of the current color enhancement, often making features that appear light on a dark background appear darker on a lighter background, or conversely. The displayed angle values, the processed image, and the point positions in the sRGB cube and ZCA space are updated accordingly. The operation is non-destructive and can be applied again to return to the previous angular configuration.

8. Split view

Clicking or tapping the **Split** button (or pressing the **T** key on desktop) divides a 2D image panel into two comparison areas. The left side displays the original reference image, while the right side shows the current processed rendering, including the active ZCA rotation, variance restoration, or channel display mode. A white vertical divider marks the boundary between the two views. The divider can be dragged horizontally with the mouse or the finger on the mobile version to adjust the relative size of the original and processed areas. When

the image is zoomed or panned, the split remains aligned with the visible image area, making local before/after comparison easier.

9. Independent component analysis (ICA).

ICA-assisted orientation is an optional automatic transformation of the whitened color space. It is computed by clicking or tapping the **ICA** button (or pressing the **I** key on desktop) from the whitened color data using the same sampled subset of pixels as the ZCA computation, with a default target of about 100,000 pixels. This subsampling keeps the computation fast enough for interactive use in the browser while preserving the main structure of the image color distribution. Once the ICA transformation has been applied, the **ICA** button appears pressed, indicating that the current view corresponds to the ICA solution. ICA does not replace manual exploration; rather, it provides an automatically computed configuration that may help separate statistically independent chromatic components. The user may then continue exploring the image manually using the rotation sliders or mouse controls. Any subsequent modification of the rotation automatically deactivates the ICA state and returns the **ICA** button to its unpressed appearance, indicating that the current configuration no longer corresponds to the ICA solution.

10. Channel visualization

Four display modes are available:

- Color: full processed RGB image.
- R: red-channel grayscale view.
- G: green-channel grayscale view.
- B: blue-channel grayscale view.

When a mode is selected, the corresponding button appears pressed.

11. ROI (Region Of Interest) processing

Clicking or tapping the **ROI** button (or pressing the **O** key on desktop) enters painting mode. In this state, the user paints directly on the image with a semi-transparent yellow brush to define the area of interest. This allows color enhancement to focus on specific chromatic elements while excluding areas whose colors differ markedly from the region under investigation. On touch-screen devices, normal one-finger image navigation is temporarily replaced by ROI painting while this mode is active.

The ROI workflow is as follows:

- Click or tap **ROI** to enter paint mode.
- On desktop, left-drag with the mouse to paint the region of interest.
- On touch-screen devices, drag one finger over the image to paint the region of interest.

- Double-click on desktop, or double-tap on touch-screen devices, to validate the ROI and recompute the ZCA transformation from the selected pixels.
- Click or tap **ROI** again to exit paint mode or to cancel an active ROI. On desktop, pressing Esc also cancels ROI mode.

When the ROI is validated, the application recomputes whitening parameters from the selected pixels of the original image. Although these parameters are estimated only from the ROI, the resulting transformation is applied to the entire image. If the painted area is too small, the operation is cancelled and a message prompts the user to paint a larger area. Once validated, the yellow overlay disappears while the ROI-derived ZCA transformation remains active. If ICA is subsequently applied, it is computed using the ROI-derived whitening transformation.

12. Presets

Presets allow you to save and instantly recall transformation states while working on an image. When a new image is loaded, Preset 1 is automatically initialized with the original transformation state, making it easy to switch back and forth between the original view and processed states. To save the current state as a preset, click or tap the **Preset** button (or press the **P** key on desktop). The application stores the current rotation values, push factor, channel mode, split-view state, and, if a ROI has been applied, the full ROI transformation. Up to five presets can be saved at a time. Each saved preset appears as a numbered button (1–5). To recall a preset, click or tap its number. The sliders and channel mode are immediately restored. If the preset includes a ROI transformation, the image is also updated accordingly. Click or tap the **X** button to clear all stored presets at once, except the first one corresponding to the original image. Presets are also automatically cleared when a new image is loaded. Note that presets are temporary: they are not saved after reloading the page, closing the browser tab, or restarting the browser.

13. Variance Restore

The **Var** slider controls how much of the original color variance is restored after whitening.

At 1.00 (default), the full inverse whitening transform is applied. The original variance structure is restored, producing the standard RASCAL rendering with more visually stable colors.

At 0.00, variance restoration is disabled. The decorrelated and variance-equalized color components are mapped directly for display, producing a stronger DStretch-like effect, with enhanced chromatic contrast and less natural colors.

Intermediate values provide a continuous transition between standard RASCAL rendering and a stronger decorrelation-stretch-like visualization.

14. Push factor

The **Push** factor is controlled by a dedicated slider, which can be used with the mouse on desktop computers or by touch interaction on mobile devices. The displayed label is updated continuously. The default state corresponds to 1.0. Values above 1.0 amplify the transformed effect, whereas values below 1.0 attenuate it.

On desktop computers, the Push factor can also be adjusted with the keyboard: **Arrow Up** increases the value, and **Arrow Down** decreases it.

15. Saving processed images

Pressing or tapping the **Export view** button exports the current processed rendering as an image file. In the WebGL version, the export is performed locally by the browser. No image data are uploaded to a server. The exported file is generated from the current working image. Therefore, if the input image was automatically resized at loading time because it exceeded the maximum working dimension, the exported image is saved at the resized working resolution rather than at the original full resolution. The output is saved as a PNG file using a suggested filename. Depending on the browser and its settings, the file may be saved directly to the default download folder or the user may be asked to choose a location, using the filename format:

`Export_<original_name>_yyyyMMdd_HHmmss.png,`

where yyyyMMdd_HHmmss corresponds to the date and time of export.

16. Limitations of the current WebGL version

Compared with the full desktop application, the WebGL version documented here does not expose several features available in the desktop version, including the processing of 3D models, interactive 3D color-cloud visualisation, radial push processing, export of multiple view panels, session save/load capabilities, and expert/basic mode switching. TIFF and RAW files cannot be processed.

17. Quick reference

Action	How to do it
Load an image	Click or tap Open image... , then select an image file from the device.
Rotate X	Move the X slider. On desktop, scroll the mouse wheel, or press Q/D . On touch-screen devices, drag one finger vertically.
Rotate Y	Move the Y slider. On desktop, right-drag on the preview canvas or press Z/S . On touch-screen devices, drag one finger horizontally.
Rotate Z	Move the Z slider. On desktop, left-drag on the preview canvas or press A/E . On touch-screen devices, use a two-finger twist gesture.
Zoom	On desktop, hold Ctrl (or ⌘ on macOS) and scroll the mouse wheel. On touch-screen devices, use a two-finger pinch gesture.
Random rotation	Click or tap Rand . On desktop, Ctrl (or ⌘ on macOS) + Shift + R can also be used. Any manual interaction immediately stops the sequence.

Compute ICA rotation	Click or tap ICA , or press I on desktop.
Adjust variance restoration	Move the Var slider.
Change push factor	Move the Push slider. On desktop, Arrow Up increases the value and Arrow Down decreases it.
Invert polarity	Click or tap Pol. +/-
Inspect one channel	Click or tap C, R, G, or B .
Activate split view	Click or tap Split , or press T on desktop. Drag the vertical divider with the mouse or with one finger to adjust the comparison area.
Paint an ROI	Click or tap ROI , then drag with the left mouse button on desktop, or drag one finger on touch-screen devices. Double-click or double-tap to process the ROI.
Cancel ROI	Click or tap ROI again. On desktop, O toggles ROI mode and Esc cancels it.
Reset all transformations	Click or tap Reset , or press R on desktop. Outside ROI mode, double-tapping the image also resets the transformation.
Store the current configuration	Click or tap Preset , or press P on desktop.
Clear presets	Click or tap × .
Save the processed image	Click or tap Export view or Export . The processed image is exported locally by the browser as a PNG file.

18. Final note

RASCAL is an exploratory visualization tool. Processed images should always be interpreted in the scientific, archaeological, or documentary context of the original data.

19. Contact

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